

# Student CRUD using LinkedList (Logic Building)

## Concepts Covered

- Duplicate checking while adding students
- Search student
- Update only if student exists
- Delete only if student exists
- Safe removeFirst() / removeLast()
- Display current students
- clear() and isEmpty()

## Java Program

```
import java.util.LinkedList;

public class StudentCRUD {

    public static void main(String[] args) {

        LinkedList<String> students = new LinkedList<>();

        // CREATE with duplicate checking
        addStudent(students, "Rahul");
        addStudent(students, "Priya");
        addStudent(students, "Amit");
        addStudent(students, "Rahul"); // duplicate

        System.out.println("\nStudents : " + students);

        students.addFirst("Principal Recommendation");
        students.addLast("Late Admission");

        System.out.println("\nAfter addFirst() and addLast() : " + students);

        if (!students.isEmpty()) {
            System.out.println("\nFirst Student : " + students.getFirst());
            System.out.println("Last Student : " + students.getLast());
        }

        searchStudent(students, "Priya");
        updateStudent(students, "Priya", "Rohan");
        deleteStudent(students, "Amit");

        if (!students.isEmpty()) {
            System.out.println("\nRemoved First : " + students.removeFirst());
        }

        if (!students.isEmpty()) {
            System.out.println("Removed Last : " + students.removeLast());
        }

        System.out.println("\nCurrent Students:");
        for (String student : students) {
            System.out.println(student);
        }

        System.out.println("\nTotal Students : " + students.size());

        students.clear();
        System.out.println("\nAfter clear() : " + students);
        System.out.println("Is List Empty? " + students.isEmpty());
    }
}
```

```

static void addStudent(LinkedList<String> students, String name) {
    if (students.contains(name)) {
        System.out.println(name + " already exists. Duplicate not allowed.");
    } else {
        students.add(name);
        System.out.println(name + " added successfully.");
    }
}

static void searchStudent(LinkedList<String> students, String name) {
    if (students.contains(name)) {
        System.out.println("\n" + name + " is enrolled.");
    } else {
        System.out.println("\n" + name + " is not enrolled.");
    }
}

static void updateStudent(LinkedList<String> students, String oldName, String
newName) {
    int index = students.indexOf(oldName);

    if (index != -1) {
        students.set(index, newName);
        System.out.println("\nUpdated " + oldName + " to " + newName);
    } else {
        System.out.println("\nCannot update. " + oldName + " not found.");
    }

    System.out.println("After Update : " + students);
}

static void deleteStudent(LinkedList<String> students, String name) {
    if (students.remove(name)) {
        System.out.println("\n" + name + " removed successfully.");
    } else {
        System.out.println("\nCannot delete. " + name + " not found.");
    }

    System.out.println("After Delete : " + students);
}

```